Chapter 3: Matrices

Practice Set A: step-wise solutions

20 questions **49** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** CBSE SQP 2025-26 — 1 mark

If for three matrices $A = [a_{ij}]_{m\times 4}$, $B = [b_{ij}]_{n\times 3}$ and $C = [c_{ij}]_{p\times q}$, the products AB and AC are both defined and are square matrices of the same order, then the values of m, n, p and q are:

(A) $m = q = 3$ **and** $n = p = 4$ **(B)** $m = 2,\ q = 3$ and $n = p = 4$

(C) $m = q = 4$ and $n = p = 3$ **(D)** $m = 4,\ p = 2$ and $n = q = 3$

Formula / idea: $P_{a\times b}\,Q_{c\times d}$ is defined only if $b = c$; the product has order $a \times d$.

1 AB is defined, so columns of A = rows of B.
$n = 4, \qquad AB$ has order $m \times 3$

2 AB is square.
$m = 3$

3 AC is defined, so columns of A = rows of C.
$p = 4, \qquad AC$ has order $m \times q = 3 \times q$

4 AC is square of the same order as AB. ✓ 1
$q = 3 \ \Rightarrow\ m = q = 3,\ n = p = 4$

5 Hence option (A).

Answer **(A)** $m = q = 3$ **and** $n = p = 4$

Watch out: Matching the rows of the first matrix with the columns of the second, which is the wrong way round.

2 MCQ CBSE 2023 1 mark

If $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$ and $A = B^2$, then x equals:

(A) ± 1 **(B)** -1 **(C)** 1 **(D)** 2

Formula / idea: Equal matrices have equal corresponding elements.

1 Find B^2.

$$B^2 = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} x^2 & 0 \\ x+1 & 1 \end{bmatrix}$$

2 Equate with A, element by element.

$$x^2 = 1 \quad \text{and} \quad x + 1 = 2$$

3 Both must hold together. ✓ 1

$x = 1$

$x = -1$ fails $x + 1 = 2$.

4 Hence option (C).

Answer **(C)** 1

Watch out: Stopping at $x^2 = 1$ and choosing ± 1; every element must match.

3 MCQ CBSE SQP 2024-25 1 mark

If $A = \begin{bmatrix} 0 & 1 & c \\ -1 & a & -b \\ 2 & 3 & 0 \end{bmatrix}$ is a skew-symmetric matrix, then the value of $a + b + c$ is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

Formula / idea: Skew-symmetric: $a_{ji} = -a_{ij}$, so every diagonal element is 0.

1 Diagonal element a_{22} must be 0.

$$a = 0$$

2 Use $a_{13} = -a_{31}$.

$$c = -2$$

3 Use $a_{23} = -a_{32}$.

$$-b = -3 \Rightarrow b = 3$$

4 Add. ✓ 1

$$a + b + c = 0 + 3 - 2 = 1$$

5 Hence option (A).

Answer **(A)** 1

Watch out: Taking $b = -3$ by forgetting the minus sign already written in $-b$.

4 **MCQ** CBSE 2023 1 mark

If $A = [a_{ij}]$ is a square matrix of order 2 such that $a_{ij} = 1$ when $i \neq j$ and $a_{ij} = 0$ when $i = j$, then A^2 is:

(A) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ **(B)** $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ **(C)** $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ **(D)** $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

1 Diagonal elements are 0, the others are 1.

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2 Multiply A by itself. ✓ 1

$$A^2 = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

3 So $A^2 = I$. Option (D).

Answer **(D)** $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Watch out: Squaring each element instead of doing row-by-column multiplication.

5 **MCQ** CBSE 2023 1 mark

If for a square matrix A, $A^2 - A + I = O$, then A^{-1} equals:

(A) A **(B)** $A + I$ **(C)** $I - A$ **(D)** $A - I$

Formula / idea: If $AB = BA = I$, then $B = A^{-1}$.

1 Rearrange the given equation.

$$A - A^2 = I$$

2 Take A common.

$$A(I - A) = I \quad \text{and} \quad (I - A)A = I$$

3 So $I - A$ is the inverse of A. ✓ 1

$$A^{-1} = I - A$$

4 Hence option (C).

Answer **(C)** $I - A$

Watch out: Writing $A^{-1} = A - I$ by losing the sign while rearranging.

6 **Assertion-Reason** Board-pattern 1 mark

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. **Assertion (A):** $(A+B)^2 = A^2 + B^2$. **Reason (R):** For these matrices A and B, $AB = BA$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $(A+B)^2 = A^2 + AB + BA + B^2$, since in general $AB \neq BA$.

1 Find AB and BA.
$AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

2 Here $AB + BA = O$, so the middle terms cancel.
$(A+B)^2 = A^2 + AB + BA + B^2 = A^2 + B^2$

3 So Assertion (A) is true.

4 But $AB \neq BA$, so Reason (R) is false.

5 A false Reason cannot explain the Assertion.

6 Hence option (C). ✓ 1

Answer **(C) Assertion (A) is true, but Reason (R) is false.**

Watch out: Thinking $(A+B)^2 = A^2 + B^2$ needs $AB = BA$; it actually needs $AB = -BA$.

7 **Short answer** PYQ-style 2 marks

Find the values of x and y if $x \begin{bmatrix} 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$.

1 Multiply by the scalars and add. ✓ ½
$\begin{bmatrix} 2x - y \\ 3x + y \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$

2	Equate corresponding elements. $2x - y = 10, \qquad 3x + y = 5$	✓ ½
3	Add the two equations. $5x = 15 \;\Rightarrow\; x = 3$	✓ ½
4	Put $x = 3$ in $3x + y = 5$. $9 + y = 5 \;\Rightarrow\; y = -4$	✓ ½
5	Hence $x = 3$ and $y = -4$.	

Answer $x = 3,\ y = -4$

Level 2

18 marks

8 **Short answer** Board-pattern 2 marks

Construct a 3×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = |2i - 3j|$.

Formula / idea: In a_{ij}, i is the row number and j is the column number.

1	Row 1 ($i = 1$). $a_{11} = |2 - 3| = 1, \qquad a_{12} = |2 - 6| = 4$	✓ ½
2	Row 2 ($i = 2$). $a_{21} = |4 - 3| = 1, \qquad a_{22} = |4 - 6| = 2$	✓ ½
3	Row 3 ($i = 3$). $a_{31} = |6 - 3| = 3, \qquad a_{32} = |6 - 6| = 0$	✓ ½
4	Write the matrix. $A = \begin{bmatrix} 1 & 4 \\ 1 & 2 \\ 3 & 0 \end{bmatrix}$	✓ ½

Answer $A = \begin{bmatrix} 1 & 4 \\ 1 & 2 \\ 3 & 0 \end{bmatrix}$

Watch out: Dropping the modulus and writing negative entries such as $a_{12} = -4$.

9 **Short answer** PYQ-style 3 marks

If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1}.

1 Find A^2. ✓ 1

$$A^2 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 9-1 & 3+2 \\ -3-2 & -1+4 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$$

2 Substitute in $A^2 - 5A + 7I$. ✓ 1

$$\begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

3 Hence $A^2 - 5A + 7I = O$.

4 Rearrange and take A common. ✓ ½

$$5A - A^2 = 7I \;\Rightarrow\; A(5I - A) = 7I \;\Rightarrow\; A^{-1} = \frac{1}{7}(5I - A)$$

$A \cdot \frac{1}{7}(5I - A) = I.$

5 Compute $5I - A$. ✓ ½

$$A^{-1} = \frac{1}{7}\begin{bmatrix} 5-3 & 0-1 \\ 0+1 & 5-2 \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

Answer $A^2 - 5A + 7I = O$; $A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

Watch out: Writing $5 - A$ instead of $5I - A$; a number cannot be subtracted from a matrix.

10 **Short answer** PYQ-style 3 marks

Express the matrix $A = \begin{bmatrix} 2 & 0 \\ 5 & 4 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

Formula / idea: $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$: the first part is symmetric, the second skew-symmetric.

1 Write the transpose. ✓ ½

$$A' = \begin{bmatrix} 2 & 5 \\ 0 & 4 \end{bmatrix}$$

2 Find the symmetric part P. ✓ ½

$$P = \frac{1}{2}(A + A') = \frac{1}{2}\begin{bmatrix} 4 & 5 \\ 5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & \frac{5}{2} \\ \frac{5}{2} & 4 \end{bmatrix}$$

3 Check that P is symmetric. ✓ ½

$$P' = \begin{bmatrix} 2 & \frac{5}{2} \\ \frac{5}{2} & 4 \end{bmatrix} = P$$

4 Find the skew-symmetric part Q.

$$Q = \frac{1}{2}(A - A') = \frac{1}{2}\begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$$

✓ ½

5 Check: $Q' = -Q$, so Q is skew-symmetric. ✓ ½

6 Add to verify.

$$P + Q = \begin{bmatrix} 2 & 0 \\ 5 & 4 \end{bmatrix} = A$$

✓ ½

Answer $A = \begin{bmatrix} 2 & \frac{5}{2} \\ \frac{5}{2} & 4 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{5}{2} \\ \frac{5}{2} & 0 \end{bmatrix}$

Watch out: Computing $\frac{1}{2}(A' - A)$ for the skew part, which flips every sign.

11 Short answer PYQ-style — 2 marks

Find a matrix A such that $2A - 3B + 5C = O$, where $B = \begin{bmatrix} -2 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 0 & -2 \\ 7 & 1 & 6 \end{bmatrix}$.

1 Make A the subject.

$$2A = 3B - 5C \Rightarrow A = \frac{1}{2}(3B - 5C)$$

✓ ½

2 Find $3B$ and $5C$.

$$3B = \begin{bmatrix} -6 & 6 & 0 \\ 9 & 3 & 12 \end{bmatrix}, \qquad 5C = \begin{bmatrix} 10 & 0 & -10 \\ 35 & 5 & 30 \end{bmatrix}$$

✓ ½

3 Subtract element by element.

$$3B - 5C = \begin{bmatrix} -16 & 6 & 10 \\ -26 & -2 & -18 \end{bmatrix}$$

✓ ½

4 Divide every element by 2. Hence the required matrix is

$$A = \begin{bmatrix} -8 & 3 & 5 \\ -13 & -1 & -9 \end{bmatrix}$$

✓ ½

Answer $A = \begin{bmatrix} -8 & 3 & 5 \\ -13 & -1 & -9 \end{bmatrix}$

Watch out: Sign slip in $0 - (-10) = 10$ in the $(1, 3)$ place.

12 **Short answer** PYQ-style 3 marks

Find x, if $\begin{bmatrix}1 & x & 1\end{bmatrix}\begin{bmatrix}1 & 3 & 2\\ 2 & 5 & 1\\ 15 & 3 & 2\end{bmatrix}\begin{bmatrix}1\\ 2\\ x\end{bmatrix} = O$.

Formula / idea: Multiply from the left: $(1\times 3)(3\times 3) = 1\times 3$, then $(1\times 3)(3\times 1) = 1\times 1$.

1 Multiply the row matrix by the 3×3 matrix. ✓ 1
$$\begin{bmatrix}1+2x+15 & 3+5x+3 & 2+x+2\end{bmatrix} = \begin{bmatrix}2x+16 & 5x+6 & x+4\end{bmatrix}$$

2 Multiply by the column matrix. ✓ 1
$$(2x+16)+2(5x+6)+x(x+4) = x^2+16x+28$$

3 Set the 1×1 matrix equal to O and factorise. ✓ ½
$$x^2+16x+28=0 \;\Rightarrow\; (x+2)(x+14)=0$$

4 Hence the values of x. ✓ ½
$$x=-2 \quad \text{or} \quad x=-14$$

Answer $x=-2$ **or** $x=-14$

Watch out: Forgetting that the last entry is multiplied by x, which loses the x^2 term.

13 **Short answer** Board-pattern 2 marks

If $A=\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}$, find $\alpha\left(0<\alpha<\frac{\pi}{2}\right)$ such that $A^2=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$.

Formula / idea: $\cos^2\alpha-\sin^2\alpha=\cos 2\alpha$ and $2\sin\alpha\cos\alpha=\sin 2\alpha$.

1 Multiply A by itself. ✓ ½
$$A^2=\begin{bmatrix}\cos^2\alpha-\sin^2\alpha & -2\sin\alpha\cos\alpha\\ 2\sin\alpha\cos\alpha & \cos^2\alpha-\sin^2\alpha\end{bmatrix}$$

2 Use the double-angle formulas. ✓ ½
$$A^2=\begin{bmatrix}\cos 2\alpha & -\sin 2\alpha\\ \sin 2\alpha & \cos 2\alpha\end{bmatrix}$$

3 Equate with the given matrix. ✓ ½
$$\cos 2\alpha=0, \qquad \sin 2\alpha=1$$

4 Since $0<2\alpha<\pi$, solve. ✓ ½
$$2\alpha=\frac{\pi}{2} \;\Rightarrow\; \alpha=\frac{\pi}{4}$$

Answer $\alpha=\frac{\pi}{4}$

Watch out: Squaring each element, giving $\cos^2\alpha$ in place of $\cos 2\alpha$.

14 **Short answer** Board-pattern 3 marks

If $A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$, verify that $(AB)' = B'A'$.

1 Find AB (order 2×2). ✓ 1

$$AB = \begin{bmatrix} 2+0+3 & 1-2+9 \\ -8+0+1 & -4+0+3 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ -7 & -1 \end{bmatrix}$$

2 Take the transpose: LHS. ✓ ½

$$(AB)' = \begin{bmatrix} 5 & -7 \\ 8 & -1 \end{bmatrix}$$

3 Write B' and A'. ✓ ½

$$B' = \begin{bmatrix} 2 & 0 & 1 \\ 1 & -1 & 3 \end{bmatrix}, \qquad A' = \begin{bmatrix} 1 & -4 \\ 2 & 0 \\ 3 & 1 \end{bmatrix}$$

4 Find $B'A'$: RHS. ✓ ½

$$B'A' = \begin{bmatrix} 2+0+3 & -8+0+1 \\ 1-2+9 & -4+0+3 \end{bmatrix} = \begin{bmatrix} 5 & -7 \\ 8 & -1 \end{bmatrix}$$

5 LHS = RHS, so $(AB)' = B'A'$ is verified. ✓ ½

Answer $(AB)' = B'A' = \begin{bmatrix} 5 & -7 \\ 8 & -1 \end{bmatrix}$, **hence verified.**

Watch out: Computing $A'B'$ instead of $B'A'$; the order reverses.

Level 3

23 marks

15 **Long answer** PYQ-style 3 marks

If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$, find the value of k such that $A^2 = kA - 2I$.

1 Find A^2. ✓ 1

$$A^2 = \begin{bmatrix} 9-8 & -6+4 \\ 12-8 & -8+4 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix}$$

2 Write $kA - 2I$. ✓ ½

$$kA - 2I = \begin{bmatrix} 3k-2 & -2k \\ 4k & -2k-2 \end{bmatrix}$$

3 Equate corresponding elements. ✓ 1

$3k - 2 = 1, \quad -2k = -2, \quad 4k = 4, \quad -2k - 2 = -4$

4 All four equations give the same value. Hence
$k = 1$ ✓ ½

Answer $k = 1$

Watch out: Writing $kA - 2$ and subtracting 2 from every element; $2I$ changes only the diagonal.

16 Long answer Board-pattern 3 marks

A factory makes three items P, Q and R. One unit of P needs 2 kg of raw material and 3 hours of labour; Q needs 1 kg and 4 hours; R needs 3 kg and 2 hours. Raw material costs ₹50 per kg and labour costs ₹40 per hour. Using matrix multiplication, find the total cost of making 20 units of P, 30 units of Q and 40 units of R.

1 Write the order (number of units) as a row matrix. ✓ ½
$N = \begin{bmatrix} 20 & 30 & 40 \end{bmatrix}$

2 Requirement matrix (rows P, Q, R; columns kg, hours) and cost column. ✓ ½
$M = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ 3 & 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 50 \\ 40 \end{bmatrix}$

3 Total raw material and labour needed: NM. ✓ 1
$NM = \begin{bmatrix} 40 + 30 + 120 & 60 + 120 + 80 \end{bmatrix} = \begin{bmatrix} 190 & 260 \end{bmatrix}$

4 Total cost $= (NM)C$. ✓ ½
$\begin{bmatrix} 190 & 260 \end{bmatrix} \begin{bmatrix} 50 \\ 40 \end{bmatrix} = 9500 + 10400 = 19900$

5 Hence the total cost is ₹19,900. ✓ ½

Answer **Total cost = ₹19,900**

Watch out: Writing the costs as a row matrix, so that the product is not defined.

17 Long answer PYQ-style 3 marks

If $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$, show that $A^2 = A$. Hence find $(A + I)^2$.

1 Row 1 of A times each column of A. ✓ ½
$\begin{bmatrix} 4 + 2 - 4 & -4 - 6 + 8 & -8 - 8 + 12 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \end{bmatrix}$

2 Row 2 of A times each column of A. ✓ ½
$\begin{bmatrix} -2 - 3 + 4 & 2 + 9 - 8 & 4 + 12 - 12 \end{bmatrix} = \begin{bmatrix} -1 & 3 & 4 \end{bmatrix}$

3 Row 3 of A times each column of A. ✓ ½
$$\begin{bmatrix}2+2-3 & -2-6+6 & -4-8+9\end{bmatrix} = \begin{bmatrix}1 & -2 & -3\end{bmatrix}$$

4 The rows of A^2 are the rows of A. ✓ ½
$$A^2 = A$$

5 Expand; A and I commute, so $AI + IA = 2A$. ✓ ½
$$(A+I)^2 = A^2 + 2A + I = A + 2A + I = 3A + I$$

6 Compute $3A + I$. ✓ ½
$$(A+I)^2 = \begin{bmatrix}7 & -6 & -12\\ -3 & 10 & 12\\ 3 & -6 & -8\end{bmatrix}$$

Answer $A^2 = A$; $(A+I)^2 = 3A + I = \begin{bmatrix}7 & -6 & -12\\ -3 & 10 & 12\\ 3 & -6 & -8\end{bmatrix}$

Watch out: Adding I (that is, 1) to every element of $3A$ instead of only the diagonal.

18 **Long answer** PYQ-style 5 marks

If $F(x) = \begin{bmatrix}\cos x & -\sin x & 0\\ \sin x & \cos x & 0\\ 0 & 0 & 1\end{bmatrix}$, show that $F(x)\,F(y) = F(x+y)$. Hence show that $[F(x)]^{-1} = F(-x)$.

Formula / idea: $\cos(x+y) = \cos x\cos y - \sin x\sin y$, $\sin(x+y) = \sin x\cos y + \cos x\sin y$.

1 Write the product. ✓ ½
$$F(x)F(y) = \begin{bmatrix}\cos x & -\sin x & 0\\ \sin x & \cos x & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos y & -\sin y & 0\\ \sin y & \cos y & 0\\ 0 & 0 & 1\end{bmatrix}$$

2 Row 1 times each column. ✓ 1
$$\begin{bmatrix}\cos x\cos y - \sin x\sin y & -\cos x\sin y - \sin x\cos y & 0\end{bmatrix}$$

3 Row 2 and row 3 times each column. ✓ 1
$$\begin{bmatrix}\sin x\cos y + \cos x\sin y & -\sin x\sin y + \cos x\cos y & 0\end{bmatrix},\quad \begin{bmatrix}0 & 0 & 1\end{bmatrix}$$

4 Use the compound-angle formulas. ✓ 1
$$F(x)F(y) = \begin{bmatrix}\cos(x+y) & -\sin(x+y) & 0\\ \sin(x+y) & \cos(x+y) & 0\\ 0 & 0 & 1\end{bmatrix}$$

5 This is F with angle $x+y$. ✓ ½
$$F(x)F(y) = F(x+y)$$

6 Put $y = -x$; similarly put $-x$ for x and x for y. ✓ ½

$F(x)F(-x) = F(0) = I, \qquad F(-x)F(x) = F(0) = I$

$F(0)$ has $\cos 0 = 1, \sin 0 = 0$, so $F(0) = I$.

7 Hence $F(-x)$ is the inverse of $F(x)$. ✓ ½

$[F(x)]^{-1} = F(-x)$

Answer $F(x)F(y) = F(x+y)$ **and** $[F(x)]^{-1} = F(-x)$

Watch out: Sign slip in the $(1,2)$ entry: it is $-(\sin x \cos y + \cos x \sin y) = -\sin(x+y)$.

19 **Long answer** PYQ-style 5 marks

If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, verify that $A^3 - 6A^2 + 9A - 4I = O$. Hence find A^{-1}.

1 Find $A^2 = A \cdot A$. ✓ 1

$$A^2 = \begin{bmatrix} 4+1+1 & -2-2-1 & 2+1+2 \\ -2-2-1 & 1+4+1 & -1-2-2 \\ 2+1+2 & -1-2-2 & 1+1+4 \end{bmatrix} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

2 Find $A^3 = A^2 \cdot A$. ✓ 1

$$A^3 = \begin{bmatrix} 12+5+5 & -6-10-5 & 6+5+10 \\ -10-6-5 & 5+12+5 & -5-6-10 \\ 10+5+6 & -5-10-6 & 5+5+12 \end{bmatrix} = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

3 Group the positive terms: $A^3 + 9A$. ✓ ½

$$A^3 + 9A = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} = \begin{bmatrix} 40 & -30 & 30 \\ -30 & 40 & -30 \\ 30 & -30 & 40 \end{bmatrix}$$

4 Group the other terms: $6A^2 + 4I$. ✓ ½

$$6A^2 + 4I = \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 40 & -30 & 30 \\ -30 & 40 & -30 \\ 30 & -30 & 40 \end{bmatrix}$$

5 The two are equal, so their difference is O. ✓ ½

$A^3 - 6A^2 + 9A - 4I = O$

6 Rearrange and take A common. ✓ ½

$A^3 - 6A^2 + 9A = 4I \;\Rightarrow\; A\left(A^2 - 6A + 9I\right) = 4I$

7 So A is invertible, with this inverse. ✓ ½

$A^{-1} = \frac{1}{4}\left(A^2 - 6A + 9I\right)$

$A \cdot \frac{1}{4}(A^2 - 6A + 9I) = I.$

8 Compute $A^2 - 6A + 9I$ and divide by 4. ✓ ½

$$A^{-1} = \frac{1}{4}\begin{bmatrix} 6-12+9 & -5+6 & 5-6 \\ -5+6 & 6-12+9 & -5+6 \\ 5-6 & -5+6 & 6-12+9 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

Answer $A^3 - 6A^2 + 9A - 4I = O$; $A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$

Watch out: Adding $9I$ to every element instead of only the diagonal when finding A^{-1}.

20 Case study Board-pattern — 4 marks

Case Study: A company makes three products x, y and z and sells them in two markets. In a year, Market I buys 10,000 units of x, 2,000 units of y and 18,000 units of z. Market II buys 6,000 units of x, 20,000 units of y and 8,000 units of z. The selling prices are ₹2.50, ₹1.50 and ₹1.00 per unit of x, y and z respectively.

(i) Write the yearly sales as a 2×3 matrix and the selling prices as a column matrix. — 1 mark

(ii) Find the revenue from Market I. — 1 mark

(iii)(a) Using matrix multiplication, find the revenue from each market and the total revenue from both markets. — 2 marks

OR

(iii)(b) If the costs of making one unit of x, y and z are ₹2.00, ₹1.00 and 50 paise respectively, find the gross profit in each market using matrix algebra. — 2 marks

(i) — 1 mark

1 Rows = markets I, II; columns = products x, y, z. ✓ ½

$$S = \begin{bmatrix} 10000 & 2000 & 18000 \\ 6000 & 20000 & 8000 \end{bmatrix}$$

2 Selling prices of x, y, z as a column matrix. ✓ ½

$$P = \begin{bmatrix} 2.50 \\ 1.50 \\ 1.00 \end{bmatrix}$$

Answer (i) $S = \begin{bmatrix} 10000 & 2000 & 18000 \\ 6000 & 20000 & 8000 \end{bmatrix}$, $P = \begin{bmatrix} 2.50 \\ 1.50 \\ 1.00 \end{bmatrix}$

(ii) — 1 mark

1 Multiply Market I's row by the price column. ✓ ½

$$\begin{bmatrix} 10000 & 2000 & 18000 \end{bmatrix}\begin{bmatrix} 2.50 \\ 1.50 \\ 1.00 \end{bmatrix} = 25000 + 3000 + 18000$$

2 Add. Revenue from Market I is ₹46,000.

$= 46000$

✓ ½

Answer (ii) ₹46,000

(iii)(a) 2 marks

1 Revenue matrix $= SP$ (order 2×3 times 3×1).

$$SP = \begin{bmatrix} 10000 & 2000 & 18000 \\ 6000 & 20000 & 8000 \end{bmatrix} \begin{bmatrix} 2.50 \\ 1.50 \\ 1.00 \end{bmatrix}$$

✓ ½

2 Multiply each row by the column.

$$= \begin{bmatrix} 25000 + 3000 + 18000 \\ 15000 + 30000 + 8000 \end{bmatrix} = \begin{bmatrix} 46000 \\ 53000 \end{bmatrix}$$

✓ 1

3 Market I ₹46,000, Market II ₹53,000. Add for the total.

$46000 + 53000 = 99000$

✓ ½

4 Hence the total revenue from both markets is ₹99,000.

Answer (iii)(a) Market I: ₹46,000; Market II: ₹53,000; total ₹99,000

OR

(iii)(b) 2 marks

1 Profit per unit = selling price − cost price.

$$\begin{bmatrix} 2.50 \\ 1.50 \\ 1.00 \end{bmatrix} - \begin{bmatrix} 2.00 \\ 1.00 \\ 0.50 \end{bmatrix} = \begin{bmatrix} 0.50 \\ 0.50 \\ 0.50 \end{bmatrix}$$

✓ ½

2 Gross profit matrix = S times the profit column.

$$\begin{bmatrix} 10000 & 2000 & 18000 \\ 6000 & 20000 & 8000 \end{bmatrix} \begin{bmatrix} 0.50 \\ 0.50 \\ 0.50 \end{bmatrix} = \begin{bmatrix} 5000 + 1000 + 9000 \\ 3000 + 10000 + 4000 \end{bmatrix}$$

✓ ½

3 Simplify.

$$= \begin{bmatrix} 15000 \\ 17000 \end{bmatrix}$$

✓ ½

4 Gross profit: Market I ₹15,000; Market II ₹17,000.

✓ ½

Answer (iii)(b) Market I: ₹15,000; Market II: ₹17,000

Watch out: Writing prices as a row matrix; it must be a column so that SP is defined.